HURWITZ THEORY

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Contents

0.1. **Surfaces.**

Definition 0.1.1. A *surface* is a two-dimensional topological manifold.

Proposition 0.1.2. (♠ TODO: mobius strip) A surface is orientable if and only if it does not contain any subset homeomorphic to a Möbius strip.

0.2. **Euler characteristic and genus.**

Definition 0.2.1. (♠ TODO: betti numbers) Let X be a topological space and let $H_n(X; \mathbb{Q})$ be the n-th singular homology group of X with coefficients in the rational numbers $\mathbb{Q}$. If the total dimension of the singular homology is finite, the *Euler characteristic of X*, denoted $\chi(X)$, is defined by the alternating sum of the Betti numbers:

$$\chi(X) = \sum_{n=0}^{\infty} (-1)^n \dim_{\mathbb{Q}} H_n(X; \mathbb{Q})$$

Definition 0.2.2. Let X be a topological space. One might denote by $g = g(X)$ the unique number in $\frac{1}{2}\mathbb{Z}$ defined by

$$g = \frac{2 - \chi(X)}{2}$$

or equivalently

$$\chi(X) = 2 - 2g$$

where $\chi(X)$ is the Euler characteristic of X defined via singular homology.

In the case that X is a surface (Definition 0.1.1), the quantity $g = g(X)$ above may be called the *genus of the surface*, especially when the surface is compact, connected, and orientable. For compact, connected, non-orientable surfaces, the genus may be called the *"Non-orientable genus"* or *"demigenus"* for emphasis. In the case that X is a disconnected surface, the above notion of genus should not be confused with the notion of total genus.

If the surface is compact, connected, and orientable, then its genus is a non-negative integer. If the surface is compact, connected, and non-orientable, then its genus is a non-negative half integer.

Theorem 0.2.3. Let S and S' be two compact, connected, orientable surfaces. S and S' have the same homeomorphism class if and only if they have the same genus (Definition 0.2.2) g. Consequently, for every non-negative integer g, there exists a unique compact, connected, orientable surface S_g, called by names such as the *closed orientable surface of genus g*, the *surface of genus g*, or the *g-fold torus*, up to homeomorphism.

1. The Burau representations

1.1. **Braid groups.**

1.1.1. *Configuration spaces.*

Definition 1.1.1. Let X be a topological space. The *ordered configuration space of n points in X*, denoted $\mathrm{Conf}_n(X)$, is the subspace of the n-fold product space X^n defined by

$$\mathrm{Conf}_n(X) = \{(x_1, \dots, x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}.$$

Equivalently, $\mathrm{Conf}_n(X) = X^n \setminus \Delta$, where $\Delta = \{(x_1, \dots, x_n) : \exists i \neq j \text{ s.t. } x_i = x_j\}$ is the fat diagonal. (♠ TODO: fat diagonal)

Definition 1.1.2. Let X be a topological space. The *unordered configuration space of n points in X*, denoted $\mathrm{UConf}_n(X)$, is the quotient space

$$\mathrm{UConf}_n(X) = \mathrm{Conf}_n(X)/S_n$$

where the symmetric group S_n acts on $\mathrm{Conf}_n(X)$ by permutation of coordinates:

$$\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$$

for $\sigma \in S_n$. The points in $\mathrm{UConf}_n(X)$ are viewed as n-element subsets of X.

Definition 1.1.3. (♠ TODO: presentation of a group) The *(Artin) braid group* B_n is the group with the presentation given by generators $\sigma_1, \dots, \sigma_{n-1}$ subject to the relations:

- $\sigma_i\sigma_j = \sigma_j\sigma_i$ for $|i-j| \geq 2$
- $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ for $1 \leq i \leq n-2$

An element of B_n may be called a *braid*. The generators $\sigma_1, \dots, \sigma_{n-1}$ may be called the *Artin generators*, *elementary braids*, *simple braids*, or *elementary crossings*.

Equivalently, B_n can be characterized as the fundamental group (Definition 1.4.3) $\pi_1(\mathrm{UConf}_n(\mathbb{D}, c_0)$ of the unordered configuration space (Definition 1.1.2) $\mathrm{UConf}_n(\mathbb{D})$ of the (unit) disk $\mathbb{D} \subseteq \mathbb{R}^2$ based at any point $c_0 \in \mathrm{UConf}_n(\mathbb{D})$.

Proposition 1.1.4. Let $\mathbb{D} \subseteq \mathbb{R}^2$ be a (closed unit) disk and let $x_1, \dots, x_n \in \mathbb{D}$ be n distinct points in the interior of $\mathbb{D}$. There is a canonical isomorphism

$$B_n \cong \mathrm{Mod}(\mathbb{D}, \partial\mathbb{D}, \{x_1, \dots, x_n\})$$

(Definition 1.1.3) where $\mathrm{Mod}(\mathbb{D}, \partial\mathbb{D}, \{x_1, \dots, x_n\})$ is the mapping class group (Definition 1.3.8) of homeomorphisms of $\mathbb{D}$ fixing the boundary $\partial\mathbb{D}$ pointwise and permuting the punctures.

Definition 1.1.5. The *pure braid group*, denoted P_n, is the kernel of the natural surjective homomorphism $\pi : B_n \to S_n$ (Definition 1.1.3) onto the symmetric group S_n, which maps a braid to the permutation of its endpoints.

Note in particular that P_n is a subset of B_n, and hence all elements of P_n are braids (Definition 1.1.3).

Topologically, P_n is canonically isomorphic to the fundamental group (Definition 1.4.3) $\pi_1(\mathrm{Conf}_n(\mathbb{D}), c_0)$ of the ordered configuration space (Definition 1.1.1) $\mathrm{Conf}_n(\mathbb{D})$ of n points in the disk $\mathbb{D} \subseteq \mathbb{R}^2$.

Configuration spaces can also be constructed for schemes.

Definition 1.1.6. Let S be a scheme and let X be a scheme over S. The *ordered configuration space of n points in X (over S)* refers to the n-fold fiber product $X^n = X \times_S X \times_S \cdots \times_S X$.

Note that it represents the functor $\mathrm{Sch}/S \to \mathbf{Sets} : T \mapsto \mathrm{Hom}_{\mathrm{Sch}/S}(T, X)^n$.

Definition 1.1.7. Let S be a scheme and let X be a scheme over S. The *configuration space of n distinct points in X (over S)* refers to the scheme

$$\mathrm{Conf}_n(X) = X^n \setminus \bigcup_{1 \leq i < j \leq n} \Delta_{ij}$$

(Definition 1.1.6) where $\Delta_{ij} \subseteq X^n = X \times_S X \times_S \cdots \times_S X$ refers to the pairwise diagonal at the i and jth coordinates — it may be constructed by first taking the diagonal $\Delta \subseteq X_i \times_S X_j = X \times_S X$ where X_i and X_j are ith and jth copies of X and by letting Δ_{ij} be the product of Δ with the copies of X except the ith and jth ones. (♠ TODO: comment on how it represents tuples of points)

This configuration space should not be confused with the unordered configuration space of n distinct points in X over S (Definition 1.1.8), which some might also denote by $\mathrm{Conf}_n(X)$

Definition 1.1.8. Let S be a scheme and let X be a scheme over S. The *unordered configuration space of n distinct points in X (over S)* refers to the scheme

$$\mathrm{UConf}_n(X) = \mathrm{Conf}_n(X)/S_n$$

(Definition 1.1.7) where $\mathrm{Conf}_n(X)$ is the ordered configuration space of n distinct points in X and S_n acts on $\mathrm{Conf}_n(X)$ by permuting coordinates.

This unordered configuration space may be denoted by $\mathrm{Conf}_n(X)$, which may overlap with notation for the ordered configuration space.

(♠ TODO: comment on how it represents tuples of points)

Definition 1.1.9. Let S be a scheme and let X be a scheme over S. For integers $n_1, \dots, n_v \geq 1$ and $n = \sum_{i=1}^{v} n_i$, the *unordered configuration space of $n_1, \dots, n_v$ distinct colored points in X (over S)* refers to the scheme

$$\mathrm{UConf}_{n_1,\dots,n_v}(X) = \mathrm{Conf}_n(X)/\prod_{i=1}^{v} S_{n_i}$$

(Definition 1.1.7) where S_{n_i} acts by permuting the n_i consecutive coordinates in the range $[n_1 + \cdots + n_{i-1} + 1, n_1 + \cdots + n_i]$. (♠ TODO: comment on how it represents tuples of points)

This configuration space may also be denoted as $\mathrm{UConf}_{n_1,\dots,n_v}(X)$.

1.2. **Defining the Burau representation.**

1.2.1. *Orientation by singular homology.*

Definition 1.2.1. (♠ TODO: generator) Let M be a topological n-manifold (with or without boundary).

A *local orientation of M at a point $x \in \operatorname{int} M$* is a choice of generator μ_x of the local homology group $H_n(M, M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}$.

Definition 1.2.2. Let M be a topological n-manifold (with or without boundary).

An *orientation of M* is a function $x \mapsto \mu_x$ assigning a local orientation (Definition 1.2.1) to each point $x \in \operatorname{int} M$ such that for every $x \in \operatorname{int} M$, there exists an open neighborhood $U \subset \operatorname{int} M$ and a class $\alpha \in H_n(M, M \setminus U; \mathbb{Z})$ such that for all $y \in U$, the map induced by the inclusion $j_y : (M, M \setminus U) \to (M, M \setminus \{y\})$ sends α to μ_y.

If M is a topological manifold with or without boundary for which an orientation exists, then M is said to be *orientable*.

A topological manifold M with or without boundary equipped with an orientation is said to be an *oriented manifold*.

Definition 1.2.3. Let M and N be topological n-manifolds (with or without boundary), equipped with orientations (Definition 1.2.2) μ^M and μ^N respectively.

A homeomorphism $f : M \to N$ is *orientation-preserving* if for every $x \in \operatorname{int} M$, the induced isomorphism on local homology

$$f_* : H_n(M, M \setminus \{x\}; \mathbb{Z}) \to H_n(N, N \setminus \{f(x)\}; \mathbb{Z})$$

maps the chosen generator μ^M_x to the chosen generator $\mu^N_{f(x)}$.

Definition 1.2.4. Let M be a topological n-manifold (with or without boundary). (♠ TODO: homeomorphism group) The *orientation-preserving homeomorphism group of M*, denoted $\mathrm{Homeo}^+(M)$, is the subgroup of the homeomorphism group $\mathrm{Homeo}(M)$ consisting of all orientation-preserving homeomorphisms (Definition 1.2.3) $M \to M$.

Definition 1.2.5. Let M be an oriented (Definition 1.2.2) topological n-manifold with or without boundary. The *group of orientation-preserving homeomorphisms fixing the boundary*, denoted $\mathrm{Homeo}^+(M, \partial M)$, is the subgroup of $\mathrm{Homeo}^+(M)$ (Definition 1.2.4) defined by

$$\mathrm{Homeo}^+(M, \partial M) = \{f \in \mathrm{Homeo}^+(M) : f(x) = x \text{ for all } x \in \partial M\}.$$

Note that this group is simply $\mathrm{Homeo}^+(M, \partial M, \emptyset)$ in the notation of Notation 1.3.7.

Theorem 1.2.6. Let M be a compact, connected, oriented (Definition 1.2.2) topological n-manifold, which may be with or without boundary. A homeomorphism $f : M \to M$ is orientation-preserving (Definition 1.2.3) if and only if the induced map $f_* : H_n(M, \partial M; \mathbb{Z}) \to H_n(M, \partial M; \mathbb{Z})$ on the relative singular homology is the identity.

1.3. Isotopy between topological spaces and between homeomorphisms of topological spaces.

Definition 1.3.1. Let M and N be topological spaces.

An *isotopy from M to N* is a homotopy $H : M \times [0, 1] \to N$ such that for each $t \in [0, 1]$, the map $h_t : M \to N$ defined by $h_t(x) = H(x, t)$ is a homeomorphism.

In practice, this terminology is used essentially exclusively in the case that M and N are topological manifolds with or without boundary.

Definition 1.3.2. Let M and N be topological spaces.

Two homeomorphisms $f, g : M \to N$ are said to be *isotopic* if there exists an isotopy $H : M \times [0, 1] \to N$ such that $h_0 = f$ and $h_1 = g$.

In practice, this terminology is used essentially exclusively in the case that M and N are topological manifolds with or without boundary.

The relation of being isotopic is an equivalence relation on the set of homeomorphisms from M to N. If $M = N$, the set of isotopy classes of homeomorphisms form a group under composition if the inverse of an isotopy is also an isotopy.

Definition 1.3.3. (♠ TODO: do more generally where isotopy might permute some points) Let M and N be topological spaces, and let $A \subset M$ be a subset. An *isotopy relative to A* is an isotopy (Definition 1.3.1) $H : M \times [0,1] \to N$ such that for all $a \in A$ and all $t \in [0,1]$, $H(a,t) = H(a,0)$. Two homeomorphisms $f, g : M \to N$ are *isotopic relative to A* if there exists such an isotopy with $h_0 = f$ and $h_1 = g$.

In practice, this terminology is used essentially exclusively in the case that M and N are topological manifolds with or without boundary and A is a submanifold of M.

Definition 1.3.4. Let M be a smooth manifold, which may be with or without boundary. A *smooth isotopy* is a smooth map $H : M \times [0,1] \to M$ such that for each $t \in [0,1]$, the map $h_t : M \to M$ is a diffeomorphism. Two diffeomorphisms $f, g : M \to M$ are *smoothly isotopic* if such a smooth isotopy exists with $h_0 = f$ and $h_1 = g$.

The relation of being isotopic defines an equivalence relation on the group of homeomorphisms $\mathrm{Homeo}(M)$. The set of equivalence classes of $\mathrm{Homeo}(M)$ under this relation is the set of path-components $\pi_0(\mathrm{Homeo}(M))$.

1.3.1. *Mapping class groups of manifolds.*

Definition 1.3.5. Let X and Y be topological spaces.

1. Write $\mathrm{Open}(Y)$ for the topology of Y and $\mathrm{Comp}(X)$ for the family of compact subsets of X. For $K \in \mathrm{Comp}(X)$ and $U \in \mathrm{Open}(Y)$, define the subset
$$\langle K, U\rangle := \{ f \in C(X,Y) \mid f(K) \subseteq U \} \subseteq C(X,Y).$$
where $C(X,Y)$ is the set of all continuous maps $X \to Y$. The collection of all such subsets is denoted by $\mathcal{S}_{\mathrm{co}}(X,Y)$.
2. The *compact-open topology on $C(X,Y)$* is the topology generated by the subbasis $\mathcal{S}_{\mathrm{co}}(X,Y)$. The resulting topological space is called the *function space from X to Y* and is commonly denoted by notations such as $\mathrm{Map}(X,Y)$, Y^X, $\mathrm{Fun}(X,Y)$, or $\mathrm{F}(X,Y)$. As a set, it equals $C(X,Y)$.
3. Let (X, x_0) and (Y, y_0) be based topological spaces. The *based function space from (X, x_0) to (Y, y_0)* is the subspace of $\mathrm{Map}(X,Y)$ consisting of all based maps f with $f(x_0) = y_0$, equivalently $C_*((X,x_0),(Y,y_0))$ endowed with the subspace topology inherited from the compact-open function space $\mathrm{Map}(X,Y)$. It itself is a based topological space whose base point is given by the constant map $X \to Y$ with value y_0. Common notations for the based function space include those which include a star or bullet such as $\mathrm{Map}_*((X,x_0),(Y,y_0))$, $\mathrm{Map}_*(X,Y)$, or Y_*^X to emphasize that the spaces involved are pointed, or those which omit a star or bullet, such as $\mathrm{Map}(X,Y)$, and Y^X.

Definition 1.3.6. Let X be a topological space. Its *homeomorphism group* is the group $\mathrm{Homeo}(X)$ consisting of (self-)homeomorphisms $X \to X$.

If X is locally compact and locally path-connected (Definition A.0.2), then $\mathrm{Homeo}(X)$ is a topological group equipped with the compact-open topology (Definition 1.3.5) (inherited from the set $C(X, X)$ of continuous maps $X \to X$).

Notation 1.3.7. Let M be a topological space. Let $A, B \subseteq M$ be subsets.

One may denote by notations roughly resembling $\mathrm{Homeo}(M, A, B)$ for the group of homeomorphisms $f : M \to M$ of M that fix A pointwise and that fix B setwise, i.e. $f|_A = \mathrm{id}_A$ and $f(B) = B$.

Moreover,

- When M is an oriented (Definition 1.2.2) manifold, one adds a superscript of $+$, e.g. $\mathrm{Homeo}^+(M, A, B)$, to refer to the subgroup of orientation preserving homeomorphisms.
- One adds a subscript of 0, e.g. $\mathrm{Homeo}_0(M, A, B)$, to refer to the subgroup of homeomorphisms $f : M \to M$ that are isotopic (Definition 1.3.3) to id_M via some isotopy $H : M \times [0, 1] \to M$ relative to A such that $H|_{B\times\{t\}} : B \to M$ fixes B setwise for all $t \in [0, 1]$.

In practice, the following are common notations that may occur:

- $\mathrm{Homeo}(M, \partial M)$ often denotes $\mathrm{Homeo}(M, \partial M, \emptyset)$, the group of homeomorphisms that fix the boundary of M pointwise. However, when M is a manifold with boundary, $\mathrm{Homeo}(M)$ may also be used to denote this very same group, which should not be confused with the group of all homeomorphisms of M.
- Letting finitely many points $p_1, \ldots, p_n \in M$, $\mathrm{Homeo}(M, \{p_1, \ldots, p_n\})$ may denote, depending on the context, either $\mathrm{Homeo}(M, \{p_1, \ldots, p_n\}, \emptyset)$ or $\mathrm{Homeo}(M, \emptyset, \{p_1, \ldots, p_n\})$, which are the groups of homeomorphisms that fix the finitely many points pointwise and setwise, respectively. In particular, elements of the latter group may permute the finitely many points.

In the case that M is a topological manifold with or without boundary, $\mathrm{Homeo}(M)$ is a topological group equipped with the compact open topology (Definition 1.3.5) inherited from the set $C(M, M)$ of continuous maps $M \to M$. In this case, whenever A is a closed subset of M and B is a finite subset of M, the subgroup $\mathrm{Homeo}(M, A, B)$ is in fact a closed subgroup of $\mathrm{Homeo}(M)$.

Definition 1.3.8. Let M be a connected, oriented (Definition 1.2.2) topological n-manifold with or without boundary. Let $A, B \subseteq M$ be subsets. Let $\mathrm{Homeo}^+(M, A, B)$ (Notation 1.3.7) be the group of orientation-preserving homeomorphisms of M that fix A pointwise and that fix B setwise.

The *(topological) mapping class group of M (with respect to A and B)*, which may be denoted $\mathrm{Mod}(M, A, B)$ or $\mathrm{Mod}_{\mathrm{Top}}(M, A, B)$, is the group of isotopy classes (Definition 1.3.2) of these homeomorphisms, defined by the quotient

$$\mathrm{Mod}(M, A, B) = \mathrm{Homeo}^+(M, A, B)/\mathrm{Homeo}_0(M, A, B)$$

where $\mathrm{Homeo}_0(M, A, B)$ is the normal subgroup consisting of homeomorphisms isotopic (Definition 1.3.3) to the identity via an isotopy fixing A pointwise throughout the isotopy and fixing B setwise throughout the isotopy.

The *(topological) mapping class group* of M without reference to explicit subsets A and B usually refers to the mapping class group $\mathrm{Mod}(M, \partial M, \emptyset)$. It may often be denoted as $\mathrm{Mod}(M)$ or as $\mathrm{Mod}(M, \partial M)$ for emphasis.

In the case that A is a closed subset and B is a finite subset, recall that $\mathrm{Homeo}^+(M, A, B)$ is a closed subgroup of $\mathrm{Homeo}^+(M)$ with the compact open topology (Definition 1.3.5). In this case, $\mathrm{Mod}(M, A, B)$ is isomorphic to the group $\pi_0(\mathrm{Homeo}^+(M, A, B))$ of path components of $\mathrm{Homeo}^+(M, A, B)$. This group of path components inherits a group structure from the topological group structure on $\mathrm{Homeo}^+(M, A, B)$.

Theorem 1.3.9. Let M be a connected, oriented (Definition 1.2.2) topological n-manifold with or without boundary. Let $A, B \subseteq M$ be subsets.

The restriction homomorphism

$$\mathrm{Mod}(M, A, B) \to \mathrm{Mod}(M \setminus B, A \setminus B, \emptyset), \quad f \mapsto f|_{M \setminus B}$$

is an isomorphism whenever

- B consists of finitely many points in the interior of M,
- M is compact,
- $\partial M \subseteq A$.

Theorem 1.3.10. Let M be a connected, oriented (Definition 1.2.2) topological n-manifold with or without boundary. Let $A, B \subseteq M$ be subsets. Let $x_0 \in A$ be a fixed basepoint of M.

1. There is a well-defined group homomorphism

$$\Phi : \mathrm{Mod}(M, A, B) \to \mathrm{Aut}(\pi_1(M, x_0))$$

(Definition 1.3.8) (Definition 1.4.3) defined by $[f] \mapsto f_*$ where f_* is the automorphism $\pi_1(M, x_0) \to \pi_1(M, x_0)$ induced by $f : (M, x_0) \to (M, x_0)$. In particular, isotopic elements of $\mathrm{Homeo}^+(M, A, B)$ (Notation 1.3.7) are sent to the same automorphism on $\pi_1(M, x_0)$. (♠ TODO: If x_0 is not fixed pointwise, then one only gets a map to the outer automorhpism group.)

2. Assuming that $x_0 \notin B$, there is a well-defined group homomorphism

$$\Phi : \mathrm{Mod}(M, A, B) \to \mathrm{Aut}(\pi_1(M \setminus B, x_0))$$

defined by $[f] \mapsto (f|_{M \setminus B})_*$. This follows from the fact that any representative f restricts to a pointed homeomorphism of the complement.

Proof.
1. (♠ TODO:)
2. Take the composition

$$\mathrm{Mod}(M, A, B) \xrightarrow{\mathrm{res}} \mathrm{Mod}(M \setminus B, A \setminus B, \emptyset) \xrightarrow{\Phi'} \mathrm{Aut}(\pi_1(M \setminus B, x_0));$$

where res is the homomorhpism induced by restricting homeomorphisms and isotopies to the complement, and Φ' is the homomorphism obtained by applying the previous part to the triple $(M \setminus B, A \setminus B, \emptyset)$.

□

(♠ TODO:)

Notation 1.3.11. Let $S^b_{g,n}$ denote a connected, orientable surface (a 2-manifold) of genus g with n punctures and b boundary components. We denote the mapping class group of this surface as $\mathrm{Mod}(S^b_{g,n})$. When $n = 0$ and $b = 0$, we write $\mathrm{Mod}(S_g)$.

(♠ TODO:)

Definition 1.3.12. Let S be an orientable surface and let c be a simple closed curve on S. A *Dehn twist about c*, denoted T_c, is an element of the mapping class group $\mathrm{Mod}(S)$ represented by a homeomorphism obtained by cutting S along c, rotating one boundary of the cut by 2π, and regluing.

(♠ TODO:)

Theorem 1.3.13. Let $S^b_{g,n}$ be a connected, orientable surface of finite type. For any integers $g, n, b \geq 0$, the mapping class group $\mathrm{Mod}(S^b_{g,n})$ is finitely generated. Specifically, if $b = 0$ and $n = 0$, the group $\mathrm{Mod}(S_g)$ is generated by a finite set of Dehn twists about a collection of simple closed curves.

(♠ TODO:)

Proposition 1.3.14. Let S be a connected, orientable surface of genus g with n punctures and b boundary components such that its Euler characteristic $\chi(S)$ is negative. The mapping class group $\mathrm{Mod}(S)$ acts on the Teichmüller space $\mathcal{T}(S)$ by precomposition. *Remark:* (♠ TODO: define Teichmüller space).

(♠ TODO:)

Theorem 1.3.15. Let S be a connected, orientable surface of finite type with $\chi(S) < 0$. The mapping class group $\mathrm{Mod}(S)$ acts properly discontinuously on the Teichmüller space $\mathcal{T}(S)$. The quotient space

$$\mathcal{M}(S) = \mathcal{T}(S)/\mathrm{Mod}(S)$$

is the moduli space of Riemann surfaces homeomorphic to S. *Remark:* (♠ TODO: define moduli space of Riemann surfaces).

Definition 1.3.16. Let M be a smooth orientable n-manifold, which may be with or without boundary. The *smooth mapping class group of M*, denoted $\mathrm{Mod}(M)$ or $\mathrm{Mod}_{\mathrm{smooth}}(M)$, not to be confused with the topological mapping class group of M (Definition 1.3.8), is the group of isotopy classes (Definition 1.3.3) of orientation-preserving (Definition 1.2.5) diffeomorphisms of M that fix the boundary ∂M pointwise:

$$\mathrm{Mod}(M) = \pi_0(\mathrm{Diff}^+(M, \partial M))$$

(Definition A.0.3)

(♠ TODO: separate notation) where $\mathrm{Diff}^+(M, \partial M)$ is the topological group of orientation-preserving diffeomorphisms restricting to the identity on ∂M, equipped with the compact-open topology.

(♠ TODO: laruent polynomial ring)

1.4. **The Burau representations.**

Definition 1.4.1. (♠ TODO: At the moment, the "reduced" version is being described. The "unreduced" version needs to be described, which should be based upon havnig $H_n(\tilde{M}, p^{-1}(x_0)$. The Hurewicz morhpism should also be discussed) Let M be a connected, oriented (Definition 1.2.2) topological n-manifold with or without boundary. Let $A, B \subseteq M$ be subsets. Let $x_0 \in A$ be a fixed basepoint of M. Let $p : (\tilde{M}, \tilde{x}_0) \to (M, x_0)$ be a pointed covering (Definition B.0.1). Write H for the subgroup of $\pi_1(M, x_0)$ corresponding to (Proposition B.0.11) the covering p.

Note that there is a "forgetful" group homomorphism

$$\mathrm{Homeo}^+(M, A, B) \to \mathrm{Aut}(M, x_0)$$

(Notation 1.3.7). Assume for all $f \in \mathrm{Homeo}^+(M, A, B)$ (Notation 1.3.7) the following "lifting criterion" holds: $f_*(H) \subseteq H$.

As a consequence of Theorem B.0.9, there is then a group homomorphism

$$\mathrm{Homeo}^+(M, A, B) \to \mathrm{Aut}_{\mathbf{Top}_*}(\tilde{M}, \tilde{x}_0).$$

For $n \geq 1$, One can show that the composition

$$\mathrm{Homeo}^+(M, A, B) \to \mathrm{Aut}_{\mathbf{Top}_*}(\tilde{M}, \tilde{x}_0) \to \mathrm{Aut}(\pi_n(\tilde{M}, \tilde{x}_0))$$

sends isotopic (Definition 1.3.3) homeomorphisms to the same elements of $\mathrm{Aut}(\pi_n(\tilde{M}, \tilde{x}_0))$ and hence the composition factors through $\mathrm{Mod}(M, A, B)$: (♠ TODO: Okay, so to get the action on H_n, I really should not go through π_n because an action on π_n is not generally sufficient to get an action on H_n. However, since H_n is functorial, I can just directly get the action on H_n to start with, and argue that H_n is "homotopy invariant", thereby obtaining a well defined map on $\mathrm{Mod}(M, A, B)$)

$$\mathrm{Mod}(M, A, B) \to \mathrm{Aut}(\pi_n(\tilde{M}, \tilde{x}_0)).$$

By the naturality of the Hurewicz homomorphism (♠ TODO:) $\pi_n(\tilde{M}, \tilde{x}_0) \to H_n(\tilde{M}, \tilde{x}_0; \mathbb{Z})$ and the functoriality of singular homology, this action induces a well-defined representation

$$\mathrm{Mod}(M, A, B) \to \mathrm{Aut}(H_n(\tilde{M}, \tilde{x}_0; \mathbb{Z})).$$

on $H_n(\tilde{M}, \tilde{x}_0; \mathbb{Z})$.

In fact, for general abelian groups G, these representations may be extended to representations

$$\mathrm{Mod}(M, A, B) \to \mathrm{Aut}(H_n(\tilde{M}, \tilde{x}_0; G))$$

as follows: by the universal coefficient theorem, there are short exact sequences

$$0 \to H_n(\tilde{M}, \tilde{x}_0; \mathbb{Z}) \otimes G \to H_n(\tilde{M}, \tilde{x}_0; G) \to \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(\tilde{M}, \tilde{x}_0; \mathbb{Z}), G) \to 0$$

natural in $(\tilde{M}, \tilde{x}_0)$. We have the group homomorphism

$$\mathrm{Homeo}^+(M, A, B) \to \mathrm{Aut}_{\mathbf{Top}_*}(\tilde{M}, \tilde{x}_0) \to \mathrm{Aut}(H_n(\tilde{M}, \tilde{x}_0; G)).$$

To see that this group homomorphism sends isotopic elements of $\mathrm{Homeo}^+(M, A, B)$ to the same automorphism on $H_n(\tilde{M}, \tilde{x}_0; G)$, the five lemma may be used on the short exact sequence.

The following names seem appropriate for the above representation:

1. *The representation of* $\mathrm{Mod}(M, A, B)$ *on the nth singular homology of the cover* $(\tilde{M}, \tilde{x}_0)$
2. A *generalized Lawrence-Krammer-Bigelow type representation*

If $G = R$ is a ring, then the representations are representations over R. (♠ TODO: representation)

Definition 1.4.2. Let $\mathbb{D} \subseteq \mathbb{R}^2$ be a (closed unit) disk, let $x_1, \ldots, x_n \in \mathbb{D}$ be n-distinct points. Let G be an abelian group (or ring). The *unreduced Burau representation*

(♠ TODO: define burau, gassner)

(♠ TODO: unreduced, reduced Burau representation)

(♠ TODO: relationship between $H_1(D)$, $H_1(D, x_0)$, $H_1(\tilde{D}, \tilde{x}_0)$, $H_1(\tilde{D}, p^{-1}(\tilde{x}_0))$)

Definition 1.4.3 (Homotopy groups of a pointed topological space)**.** For any pointed topological space (X, x_0) and integer $n \geq 0$, the *n-th homotopy set of X at x_0*, denoted $\pi_n(X, x_0)$ or simply $\pi_n(X)$ when the base point x_0 is understood or a choice of base point does not really matter, is defined as the set of all homotopy classes (rel. ∂I^n) of paired maps

$$f : (I^n, \partial I^n) \to (X, \{x_0\}),$$

where $I^n = [0, 1]^n$ and ∂I^0 is taken to be the singleton I^0 instead of the literal boundary thereof. Equivalently, one can also define it as the set of all homotopy classes of pointed maps

$$f : (S^n, s_0) \to (X, x_0),$$

where S^n is the n-sphere (♠ TODO:) and $s_0 \in S^n$ is a choice of base point. In other words, $\pi_n(X, x_0)$ is by definition $[(S^n, s_0), (X, x_0)]_*$.

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of pointed maps, and for $n \geq 2$, it is abelian. As such, for $n \geq 1$, it is appropriate to refer to $\pi_n(X, x_0)$ as the *nth homotopy group of (X, x_0)*.

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops $\gamma : [0, 1] \to X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

For every $n \geq 0$, the assignment $(X, x_0) \mapsto \pi_n(X, x_0)$ is functorial — in fact

1. For $n = 0$, it is a functor $\mathrm{Top}_* \to \mathbf{Sets}$,

2. For $n = 1$, it is a functor $\mathrm{Top}_* \to \mathbf{Grp}$,
3. For $n \geq 2$, it is a functor $\mathrm{Top}_* \to \mathbf{Ab}()$.

Given a map $f : (X, x_0) \to (Y, y_0)$ of pointed spaces, the induced map $\pi_n(X, x_0) \to \pi_n(Y, y_0)$ is often denoted by f_*.

We further note that the set $\pi_n(X, x_0) = [(S^n, s_0), (X, x_0)]_*$ is identifiable with $\pi_0(\mathrm{Map}_*((S^n, s_0), (X, x_0)))$.

2. Hurwitz spaces

2.1. Topological Hurwitz space.

2.2. Algebro-geometric Hurwitz space.

2.2.1. *Nielsen classes.*

Definition 2.2.1. Let G be a group. The *Schur multiplier of G* is the second group homology $H_2(G, \mathbb{Z})$ where the action of G on $\mathbb{Z}$ is trivial.

By the universal coefficient theorem (♠ TODO: state a univeresal coefficient theorem for group homology/cohomology), there is an isomorphism

$$H^2(G, \mathbb{C}^*) \cong \mathrm{Hom}(H_2(G, \mathbb{Z}), \mathbb{C}^*),$$

i.e. $H^2(G, \mathbb{C}^*)$ is the Pontryagin dual of $H_2(G, \mathbb{Z})$. (♠ TODO: pontryagin dual applicable here)

Theorem 2.2.2 (see [?, 6.8.8])**.** Let G be a group. Let F be a free group and let $R \triangleleft F$ be a normal subgroup such that $G \cong F/R$. There is an isomorphism

$$H_2(G, \mathbb{Z}) \cong \frac{R \cap [F, F]}{[F, R]}$$

(♠ TODO: derived subgroup)

Definition 2.2.3. Let G be a group.

Represent the Schur multiplier (Definition 2.2.1) $H_2(G, \mathbb{Z})$ (♠ TODO:)

We say that is *generated by commutators* if $H_2(G, \mathbb{Z}) \cap \{g^{-1}h^{-1}gh | g, h \in R\}$ generates M

Theorem 2.2.4 ([?, Theorem, Appendix A])**.** Let G be a finite group. Suppose that the Schur multiplier (Definition 2.2.1) $H_2(G, \mathbb{Z})$ is generated by commutators (Definition 2.2.3)

There exists an integer N with the following property: If $\sigma = (\sigma_1, \ldots, \sigma_r) \in G^r$ satisfies $n_C(\sigma) \geq N$ (♠ TODO: $n_c(\sigma)$) for all conjugacy classes $C \neq \{1\}$ of G, the nthe Hurwitz class of σ (♠ TODO: hurwitz class) consists of all $\tau = (\tau_1, \ldots, \tau_r) \in G^r$ with $\tau_1, \ldots, \tau_r = \sigma_1, \ldots, \sigma_r$ and $S(\tau) = S(\sigma)$ (♠ TODO: $S(\tau)$)

2.2.2. *Racks and quandles.*

Definition 2.2.5. A *rack* is a set c equipped with an *action map*

$$\triangleright : c \times c \to c, \quad (a, b) \mapsto a \triangleright b$$

satisfying the following equivalent definitions:

1. for all $n \geq 1$ and all $1 \leq i \leq n-1$, the operation
$$\sigma_i : c^n \to c^n$$
$$(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, x_{i+2}, \ldots, x_n) \mapsto (x_1, \ldots, x_{i-1}, x_{i+1}, x_{i+1} \triangleright x_i, x_{i+2}, \ldots, x_n)$$
defines an action of the braid group B_n (Definition 1.1.3) where $\sigma_i \in B_n$ is the ith elementary braid.
2. the map $x \triangleright (-) : c \to c$ is a bijection for each $x \in c$ and $x \triangleright (y \triangleright z) = (x \triangleright y) \triangleright (x \triangleright z)$.

Definition 2.2.6. Let $(X, \triangleright)$ and $(Y, \triangleright')$ be racks (Definition 2.2.5). A *homomorphism of racks* is a map $f : X \to Y$ such that for all $x_1, x_2 \in X$, the condition $f(x_1 \triangleright x_2) = f(x_1) \triangleright' f(x_2)$ holds.

Definition 2.2.7. The *category of racks* has as objects racks (Definition 2.2.5) and as morphisms the homomorphisms between racks (Definition 2.2.6). This category may be denoted **Rack**.

Definition 2.2.8. Let $(X, \triangleright)$ be a rack (Definition 2.2.5). For each $x \in X$, the *left translation map* $L_x : X \to X$, defined by $L_x(y) = x \triangleright y$, is an automorphism of the rack (Definition 2.2.6). If $(X, \triangleright)$ is a quandle (Definition 2.2.14), then the idempotency axiom implies that x is a fixed point of L_x for all $x \in X$.

Definition 2.2.9. Let G be a group and $S \subset G$ a subset that is invariant under conjugation (Definition C.0.1), such that for all $g \in G$ and $s \in S$, $gsg^{-1} \in S$. The *rack of a conjugacy invariant subset* is the rack (Definition 2.2.5) with underlying set S equipped with the operation $x \triangleright y = xyx^{-1}$ for all $x, y \in S$.

In the case that $S = G$, one might call this rack the *conjugation rack* or *conjugation quandle* (Definition 2.2.14) of G and denote by $\mathrm{Conj}(G)$. Note that the assignment of a group to its conjugation rack is functorial from the category of groups to the category of racks (Definition 2.2.7).

Definition 2.2.10. Let $(X, \triangleright)$ be a rack (Definition 2.2.5). The *automorphism group of the rack*, denoted $\mathrm{Aut}(X)$, is the set of all bijections $f : X \to X$ such that $f(x \triangleright y) = f(x) \triangleright f(y)$ for all $x, y \in X$, forming a group under composition.

Definition 2.2.11. Let $(X, \triangleright)$ be a rack (Definition 2.2.5). The *inner automorphism group* $\mathrm{Inn}(X)$ is the subgroup of the permutation group $\mathrm{Sym}(X)$ generated by the left translations (Definition 2.2.8) $L_x : X \to X$ defined by $L_x(y) = x \triangleright y$ for all $x, y \in X$.

Definition 2.2.12. Let $(X, \triangleright)$ be a rack (Definition 2.2.5) and let $F(X)$ be the free group generated by the set X. The *reduced structure group of a rack*, often denoted G_X, is the quotient of $F(X)$ by the normal subgroup generated by the relations $x \triangleright y = xyx^{-1}$ for all $x, y \in X$. (♠ TODO: normal subgroup generated by)

The reduced structure group may also be called the *adjoint group* or *structure group*.

Proposition 2.2.13. Let $(X, \triangleright)$ be a rack (Definition 2.2.5) and let G_X be its reduced structure group (Definition 2.2.12). There exists a unique rack homomorphism (Definition 2.2.6) $\phi : X \to G_X$ where G_X is viewed as a rack (Definition 2.2.9) under conjugation, such that for any group H and any rack homomorphism $f : X \to H$, there exists a unique group homomorphism $\bar{f} : G_X \to H$ satisfying $f = \bar{f} \circ \phi$.

In other words, the functor from the category of racks (Definition 2.2.7) to the category of groups sending a rack X to G_X is left adjoint to the functor sending a group to its conjugacy rack (Definition 2.2.9).

Definition 2.2.14. A *quandle* is a rack (Definition 2.2.5) $(X, \triangleright)$ that satisfies the idempotency axiom: for all $x \in X$, $x \triangleright x = x$.

Definition 2.2.15. Let $(X, \triangleright)$ and $(Y, \triangleright')$ be quandles (Definition 2.2.14). A *homomorphism of quandles* is a map $f : X \to Y$ that is a homomorphism of the underlying racks (Definition 2.2.6), i.e. a map satisfying $f(x_1 \triangleright x_2) = f(x_1) \triangleright' f(x_2)$ for all $x_1, x_2 \in X$.

Definition 2.2.16. The *category of quandles*, denoted **Qnd**, is the full subcategory of the category of racks (Definition 2.2.7) whose objects are the quandles (Definition 2.2.14).

Definition 2.2.17. (♠ TODO: quotient of a set by equivalence relation) (♠ TODO: equivalence generated by relations) Given a rack (Definition 2.2.5) $(X, \triangleright)$, its *quandle-ization* is the quandle (Definition 2.2.14) on $X/\sim$ whose action map (Definition 2.2.5) is inherited from $\triangleright$, where $\sim$ is the congruence on X generated by the relations $X \triangleright x \sim x$ for all $x \in X$. The assignment sending a rack to its quandle-ization is functorial.

Proposition 2.2.18. The inclusion functor ι : **Qnd** $\hookrightarrow$ **Rack**(**Definition 2.2.5**) is right adjoint to the quandle-ization functor (Definition 2.2.17) Q : **Rack** $\to$ **Qnd**.

Definition 2.2.19. Let c be a rack (Definition 2.2.5). There is a rack whose underlying set is the orbit of c under the action of the reduced structure group (Definition 2.2.12) of c and whose action map is given by $x \triangleright y = y$ for all $x, y \in c/c$. We may refer to this rack c/c as the *set of components of the rack c*, the *set of orbits of the rack c*, the *orbit rack of c*, or the *constant rack of c*.

3. Lifting invariants

4. Homological stability of Hurwitz spaces

Appendix A. Connectivity of topological spaces

Definition A.0.1. Let X be a topological space. It is said to be *path connected* if for any two points x_0, x_1 in X, there is some path $\gamma : [0,1] \to X$ in X such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

Definition A.0.2. Let X be a topological space. It is said to be *locally path connected* if for any $x \in X$, there is an open neighborhood $x \in U \subseteq X$ that is path connected (Definition A.0.1).

Definition A.0.3. Let X be a topological space.

Let $x \in X$. The *path component of x in X* is the maximal path connected subspace of X containing x.

The set of path components in X is denoted $\pi_0(X)$. For any $x \in X$, $\pi_0(X)$ coincides, as a set, with the zeroth homotopy set (Definition 1.4.3) $\pi_0(X, x_0)$ of the pointed space (X, x_0).

Definition A.0.4. A topological space X is *simply connected* if it is path-connected (Definition A.0.1) and its fundamental group (Definition 1.4.3) $\pi_1(X, x_0)$ is trivial for some (and hence every) base point $x_0 \in X$. Equivalently, X is simply connected if any two continuous maps $f, g : [0,1] \to X$ with $f(0) = g(0)$ and $f(1) = g(1)$ are homotopic relative to $\{0,1\}$.

Proposition A.0.5. Let X be a topological space. X is simply connected (Definition A.0.4) if and only if it is path-connected (Definition A.0.1) and every continuous map $S^1 \to X$ from the unit circle into X is null-homotopic.

Definition A.0.6. A topological space X is *locally simply connected* if it admits a basis of open sets $\{U_\alpha\}$ such that each U_α is simply connected. That is, for every $x \in X$ and every open neighborhood V of x, there exists a simply connected open neighborhood $U \subset V$ of x.

Proposition A.0.7. Let X be a topological space. If X is locally simply connected (Definition A.0.6), then X is both locally path-connected (Definition A.0.2) and semi-locally simply connected (Definition A.0.10).

Proposition A.0.8. Every topological manifold, with or without boundary, is locally simply connected (Definition A.0.6) because every point has a neighborhood homeomorphic to an open ball in $\mathbb{R}^n$ or $\mathbb{H}^n$, both of which are simply connected.

Definition A.0.9. (♠ TODO: contractible) Let X be a topological space. X is *locally contractible* if every point $x \in X$ has a basis of contractible open neighborhoods. Since every contractible space is simply connected, every locally contractible space is locally simply connected (Definition A.0.6). (♠ TODO: simply connected space)

Definition A.0.10. A topological space X is *semi-locally simply connected* if for every point $x \in X$, there exists an open neighborhood U of x such that the inclusion-induced homomorphism on fundamental groups (Definition 1.4.3)

$$i_* : \pi_1(U, x) \to \pi_1(X, x)$$

is the trivial homomorphism.

Proposition A.0.11. Let X be a topological space. The condition of semi-local simple connectivity (Definition A.0.10) is equivalent to the requirement that every point $x \in X$ has an open neighborhood U such that every loop in U based at x is null-homotopic in X. Note that the loops are not required to be null-homotopic within U itself. (♠ TODO: null-homotopic)

Appendix B. Covering spaces

Definition B.0.1. 1. Let X be a topological space.

A *covering space of X* is a pair $(\tilde{X}, p)$ where $\tilde{X}$ is a topological space and $p : \tilde{X} \to X$ is a continuous surjective map such that for every $x \in X$, there exists an open neighborhood U of x (called an *evenly covered neighborhood)* for which the preimage $p^{-1}(U)$ is a disjoint union of open sets $V_i \subset \tilde{X}$, each of which is mapped homeomorphically onto U by the restriction $p|_{V_i}$.

2. Let (X, x_0) be a pointed topological space. A *(pointed) covering of (X, x_0)* is a pointed topological space $(\tilde{X}, \tilde{x}_0)$ along with a covering $p : \tilde{X} \to X$ of X such that p is a map of pointed space.

Definition B.0.2. 1. Let X be a topological space.

A *morphism of covering spaces of X from $p_1 : \tilde{X}_1 \to X$ to $p_2 : \tilde{X}_2 \to X$* is a continuous map $f : \tilde{X}_1 \to \tilde{X}_2$ such that $p_2 \circ f = p_1$.

2. Let (X, x_0) be a pointed topological space. Let $p_1 : (\tilde{X}_1, \tilde{x}_1) \to (X, x_0)$ to $p_2 : (\tilde{X}_2, \tilde{x}_2) \to (X, x_0)$ be pointed coverings (Definition B.0.1) of (X, x_0). A *morphism $f : p_1 \to p_2$ of the pointed coverings* is a pointed map $f : (\tilde{X}_1, \tilde{x}_1) \to (\tilde{X}_2, x_2)$ such that $p_2 \circ f = p_1$ and $f(\tilde{x}_1) = \tilde{x}_2$.

Definition B.0.3. Let $p : \tilde{X} \to X$ be a covering space (Definition B.0.1) of a topological space X. A *deck transformation* (or *covering transformation*, or *automorphism*) of p is a homeomorphism $f : \tilde{X} \to \tilde{X}$ such that $p \circ f = p$. The set of all deck transformations forms a group under composition, denoted $\mathrm{Aut}(p)$ or $\mathrm{Deck}(p)$.

Definition B.0.4. Let X be a topological space. A *universal cover of X* is a pair $(\tilde{X}, p)$ where $p : \tilde{X} \to X$ is a covering space (Definition B.0.1) such that $\tilde{X}$ is simply connected (Definition A.0.4).

Theorem B.0.5. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space (Definition B.0.1) of a path-connected (Definition A.0.1) and locally path-connected space (Definition A.0.2) X. The induced map on fundamental groups (Definition 1.4.3) $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$ is an injective group homomorphism.

Definition B.0.6. Let $p : \tilde{X} \to X$ be a covering space (Definition B.0.1) of a topological space X. Let $f : Y \to X$ be a continuous map of topological spaces.

A *lift of f (via p)* is a continuous map $\tilde{f} : Y \to \tilde{X}$ satisfying $p \circ \tilde{f} = f$.

Similarly, for a pointed covering $p : (\tilde{X}, x_0) \to (X, x_0)$ and a pointed morphism $f : (Y, y_0) \to (x, x_0)$, a *lift of f (via p)* is a pointed map $\tilde{f} : (Y, y_0) \to (\tilde{X}, x_0)$ satisfying $p \circ \tilde{f} = f$.

Definition B.0.7. A covering space (Definition B.0.1) $p : \tilde{X} \to X$ of a topological space X is *normal* (or *Galois)* if for every $\tilde{x}_0 \in X_0$, the image of the induced map $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(\tilde{X}, p(\tilde{x}_0))$ on fundamental groups (Definition 1.4.3) is the normal subgroup of $\pi_1(\tilde{X}, p(\tilde{x}_0))$

Theorem B.0.8. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space (Definition B.0.1) and let $f : (Y, y_0) \to (X, x_0)$ be a continuous map where Y is path-connected (Definition A.0.1) and locally path-connected (Definition A.0.2). A lift $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x}_0)$ exists if and only if the image of the fundamental group (Definition 1.4.3) of (Y, y_0) is contained in the image of the fundamental group of $(\tilde{X}, \tilde{x}_0)$, i.e.

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

If such a lift exists, it is unique.

See also Theorem B.0.9.

Theorem B.0.9. Let $p_X : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ and $p_Y : (\tilde{Y}, \tilde{y}_0) \to (Y, y_0)$ be covering spaces of pointed topological spaces. Let $f : (X, x_0) \to (Y, y_0)$ be a continuous map, and assume $\tilde{X}$ is path-connected (Definition A.0.1) and locally path-connected (Definition A.0.2).

There exists a unique continuous lift $\tilde{f} : (\tilde{X}, \tilde{x}_0) \to (\tilde{Y}, \tilde{y}_0)$ such that $p_Y \circ \tilde{f} = f \circ p_X$ if and only if

$$f_*(p_{X*}(\pi_1(\tilde{X}, \tilde{x}_0))) \subseteq p_{Y*}(\pi_1(\tilde{Y}, \tilde{y}_0))$$

In the specific case where f is the identity on X, this provides the necessary and sufficient condition for the existence of a morphism of covering spaces (pointed or unpointed). Any such morphism between connected covering spaces is itself a covering map. See also Theorem B.0.8.

Theorem B.0.10. Let X be a connected, locally path-connected (Definition A.0.2), and semi-locally simply connected (Definition A.0.10) space with base point x_0. The universal cover (Definition B.0.4) $\tilde{X}$ can be constructed as the space of homotopy classes of paths starting at x_0:

$$\tilde{X} = \{[\gamma] \mid \gamma : [0, 1] \to X, \gamma(0) = x_0\}$$

with the covering map $p : \tilde{X} \to X$ given by $p([\gamma]) = \gamma(1)$. For any connected covering space (Definition B.0.1) $q : Y \to X$, there exists a covering map $r : \tilde{X} \to Y$ such that $q \circ r = p$.

Proposition B.0.11. Let X be a connected, locally path-connected (Definition A.0.2), and semi-locally simply connected (Definition A.0.10) space with base point x_0. There are bijective correspondences:

1. Between the set of isomorphism classes of connected pointed covering spaces (Definition B.0.1) $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ and the set of subgroups of $\pi_1(X, x_0)$, via $((\tilde{X}, \tilde{x}_0), p) \mapsto p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.
2. Between the set of isomorphism classes of connected covering spaces $p : \tilde{X} \to X$ and the set of conjugacy classes of subgroups of $\pi_1(X, x_0)$, via $p \mapsto [p_*(\pi_1(\tilde{X}, \tilde{x}_0))]$. (♠ TODO: conjugacy class)

Theorem B.0.12. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering with $\tilde{X}$ (and hence X) connected. The group of deck transformations (Definition B.0.3) $\mathrm{Aut}(p)$ is isomorphic to the quotient $N(H)/H$, where $H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ and $N(H)$ is its normalizer in $\pi_1(X, x_0)$. If the covering is normal (Definition B.0.7) (i.e., H is a normal subgroup of $\pi_1(X, x_0)$), then:

$$\mathrm{Aut}(p) \cong \pi_1(X, x_0)/p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

For the universal cover, $\mathrm{Aut}(p) \cong \pi_1(X, x_0)$.

Appendix C. Miscellaneous definitions

Definition C.0.1. Let G be a group. A subset $S \subset G$ is said to be *conjugacy invariant* or *invariant under conjugation by G* if for all $g \in G$ and all $s \in G$, the conjugate gSg^{-1} is an element of S.